

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seven Semester of B. Tech. (EE) Examination

Dec 2012

EE404 Energy Management and Conservation

Date: 06.12.2012, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (1) The year in which India ratified the Kyoto protocol _____ [07]

(2) The primary energy consumption of India is _____

(3) Which instrument is used to monitor O_2 , CO in flue gas _____

(4) In cogeneration, the system efficiencies can go up to _____ %

(5) Infrared thermometer is used to measure _____

(6) Lux meter is used to measure _____

(7) Which country has the largest share of the global coal reserves _____

Q - 2 (a) Explain Need for Electrical Load Management and Step By Step Approach for Maximum Demand Control. [07]

(b) Explain Indian Energy Scenario. [07]

OR

Q - 2 Write Short notes: (Any Two) [14]

- (1) Energy Efficient Motors
- (2) Benchmarking
- (3) Energy Conservation in Transportation

Q - 3 (a) Describe Performance evaluation of Typical Furnace With Sankey Diagram. [07]

(b) What is Energy Management and Write Seven Principal of Energy Management. [07]

OR

- Q - 3 (a) Explain Role of Energy Management Cell & Objective of Energy Management Cell. [07]
(b) Explain Energy Conservation in Paper Industry. [07]

SECTION - II

- Q - 4 (a) Write the Points of Electrical System Optimization & Describe any Two in Details. [07]
Q - 5 (a) Write the Potential Energy Conservation Opportunities in Lighting system. [07]
(b) Explain in details Classification of Cogeneration System [07]

OR

- Q - 5 (a) Write the Potential Energy Conservation Opportunities in Motor and Transformer. [07]
(b) Explain Energy Staffing, planning and Energy Monitoring. [03]
(c) From Energy Conservation Act, 2001 list five Important duties of an Energy manager. [04]
Q - 6 (a) Write Need for Energy Audit and Explain Detailed Energy Audit Methodology. [07]
(b) Explain Energy Audit of Illumination System & Electrical System. [07]

OR

- Q - 6 (a) Write Benefits of Energy Management. [07]
(b) Write Types of Pumped-Hydro Schemes & Explain Underground Pumped Hydro Energy Storage. [07]
